You

Which One Earth bioregion contains the point at longitude −122.3321 and latitude 47.6062? Use the boundary geometry stored in the repository, and show the SPARQL you execute.

GraphTalker

The point at longitude **-122.3321**, latitude **47.6062** (Seattle, WA) falls within:

Property	Value
Bioregion	**Pacific Northwest Coastal Forests**
Identifier	**NA15**
URI	`https://www.in2use.com/one_earth_bioregions/Pacific_Northwest_Coastal_`

How the query works:

The query uses the GeoSPARQL function `geof:sfContains` to test whether each bioregion's stored boundary polygon spatially contains the given point. Specifically:

1. It finds all instances of `oeb:Bioregion` that have a linked `geo:Geometry` with a WKT serialization (`geo:asWKT`).
2. It applies `geof:sfContains(?wkt, "POINT(-122.3321 47.6062)"^^geo:wktLiteral)` in a FILTER -- this returns true only when the bioregion's polygon **contains** the specified point.
3. The result is the **Pacific Northwest Coastal Forests (NA15)** bioregion.

(The two duplicate rows are because the bioregion has two label/identifier bindings that both match; the result is a single bioregion.)

Would you like me to store this query in the Query Library for future reuse?

You

Yes, please save that query

GraphTalker

Done! The query has been saved to the Query Library with the title **"Find bioregion containing a geographic point"**. You can find it in the Query Library tab. To reuse it for a different location, just swap out the longitude and latitude values in the `POINT(...)` literal.

You

For longitude −122.3321 and latitude 47.6062, identify the containing bioregion, its subrealm, and its realm. Use the stored geometry and the ontology's region relationships in one SPARQL query. Return one distinct row and show the query.

GraphTalker

One distinct row, as requested. Here are the results:

Level	Name
Bioregion	Pacific Northwest Coastal Forests (NA15)
Subrealm	North Pacific Coast
Realm	Northern America

How the query works:

The query combines a geospatial containment test with the ontology's region hierarchy in a single traversal:

```
Point(-122.3321 47.6062)
        |
        | geof:sfContains(polygon, point)
        v
   Bioregion (Pacific Northwest Coastal Forests, NA15)
        |
        | oeb:is_direct_region_part_of
        v
   Subrealm (North Pacific Coast)
        |
        | oeb:is_direct_region_part_of
        v
   Realm (Northern America)
```

1. **Spatial filter** -- Each `oeb:Bioregion` has a linked `geo:Geometry` with a WKT polygon. The `geof:sfContains` function tests which polygon contains the given point.
2. **Hierarchy walk** -- The predicate `oeb:is_direct_region_part_of` provides the direct parent link. The query walks it twice: Bioregion to Subrealm (typed `oeb:Subrealm`), then Subrealm to Realm (typed `oeb:Realm`).
3. **DISTINCT** eliminates any duplicate rows from multiple label or type bindings.

Would you like me to store this query in the Query Library?

You

Yes, please save to the Query Library.

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Saved! The query is now stored in the Query Library with the title **"Find bioregion, subrealm, and realm containing a geographic point"**. You can find it in the Query Library tab.

You

Which other bioregions share a boundary with Pacific Northwest Coastal Forests (NA15)? Calculate this from their stored geometries. Return each name and identifier once.

Here are the **5 bioregions that share a boundary** with Pacific Northwest Coastal Forests (NA15):

Identifier	Bioregion Name
NA5	Far Northern Pacific Coast
NA13	Greater Rockies And Mountain Forests
NA14	Northwest Intermountain Conifer Forests
NA16	Cascades Mountain Forests And Valleys
NA31	Greater California

Here is the SPARQL query that was executed:

```
PREFIX geo:     <http://www.opengis.net/ont/geosparql#>
PREFIX geof:    <http://www.opengis.net/def/function/geosparql/>
PREFIX oeb:     <https://www.in2use.com/one_earth_bioregions/>
PREFIX rdfs:    <http://www.w3.org/2000/01/rdf-schema#>
PREFIX dcterms: <http://purl.org/dc/terms/>

SELECT DISTINCT ?neighborLabel ?neighborId WHERE {
  # Get the reference bioregion's geometry
  ?na15 a oeb:Bioregion ;
        dcterms:identifier "NA15" ;
        geo:hasGeometry ?na15Geom .
  ?na15Geom geo:asWKT ?na15Wkt .

  # Find other bioregions whose geometry touches it
  ?neighbor a oeb:Bioregion ;
            rdfs:label ?neighborLabel ;
            dcterms:identifier ?neighborId ;
            geo:hasGeometry ?nGeom .
  ?nGeom geo:asWKT ?nWkt .

  FILTER (?neighbor != ?na15)
  FILTER (geof:sfTouches(?na15Wkt, ?nWkt))
}
ORDER BY ?neighborId
```

How it works:

```
  NA15 (Pacific Northwest Coastal Forests)
    |
    | geo:hasGeometry --> Geometry --> geo:asWKT --> WKT polygon
    |
    | geof:sfTouches(na15Wkt, neighborWkt)
    |
    +--> NA5  Far Northern Pacific Coast
    +--> NA13 Greater Rockies And Mountain Forests
    +--> NA14 Northwest Intermountain Conifer Forests
    +--> NA16 Cascades Mountain Forests And Valleys
```

```
  +--> NA31 Greater California
```

1. The query retrieves the WKT polygon for NA15 using its `dcterms:identifier`.
2. It then iterates over all other `oeb:Bioregion` instances and retrieves their WKT polygons.
3. `geof:sfTouches` returns true when two geometries share boundary points but their interiors do not overlap -- the precise definition of "sharing a boundary."
4. `DISTINCT` and the `FILTER (?neighbor != ?na15)` ensure each neighbor appears exactly once and NA15 itself is excluded.

Would you like me to store this query in the Query Library?

You

Which of those neighboring bioregions belong to a different realm from Pacific Northwest Coastal Forests? Show the bioregion names and their realms.

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Of the five neighbors, only **one** belongs to a different realm:

Bioregion	ID	Realm
Far Northern Pacific Coast	NA5	**Subarctic America**

The other four neighbors (NA13, NA14, NA16, NA31) all belong to the same realm as Pacific Northwest Coastal Forests -- **Northern America** -- so they are excluded.

Here is the SPARQL query:

```
PREFIX geo:     <http://www.opengis.net/ont/geosparql#>
PREFIX geof:    <http://www.opengis.net/def/function/geosparql/>
PREFIX oeb:     <https://www.in2use.com/one_earth_bioregions/>
PREFIX rdfs:    <http://www.w3.org/2000/01/rdf-schema#>
PREFIX dcterms: <http://purl.org/dc/terms/>

SELECT DISTINCT ?neighborLabel ?neighborId ?neighborRealmLabel WHERE {
  # Get the reference bioregion (NA15) and its realm
  ?na15 a oeb:Bioregion ;
        dcterms:identifier "NA15" ;
        geo:hasGeometry ?na15Geom .
  ?na15Geom geo:asWKT ?na15Wkt .

  ?na15 oeb:is_direct_region_part_of ?na15Subrealm .
  ?na15Subrealm a oeb:Subrealm ;
                oeb:is_direct_region_part_of ?na15Realm .
  ?na15Realm a oeb:Realm .

  # Find neighboring bioregions that touch NA15
  ?neighbor a oeb:Bioregion ;
            rdfs:label ?neighborLabel ;
            dcterms:identifier ?neighborId ;
            geo:hasGeometry ?nGeom .
  ?nGeom geo:asWKT ?nWkt .
```

```
  ?nGeom geo:asWKT ?nWkt .

  FILTER (?neighbor != ?na15)
  FILTER (geof:sfTouches(?na15Wkt, ?nWkt))

  # Get the neighbor's realm
  ?neighbor oeb:is_direct_region_part_of ?nSubrealm .
  ?nSubrealm a oeb:Subrealm ;
             oeb:is_direct_region_part_of ?neighborRealm .
  ?neighborRealm a oeb:Realm ;
                 rdfs:label ?neighborRealmLabel .

  # Keep only neighbors in a DIFFERENT realm
  FILTER (?neighborRealm != ?na15Realm)
}
ORDER BY ?neighborId
```

How it works:

```
NA15 (Pacific Northwest Coastal Forests)
  |
  +--[geo:hasGeometry]--> WKT polygon
  |        |
  |        | geof:sfTouches
  |        v
  |   Neighbor bioregion --> WKT polygon
  |
  +--[is_direct_region_part_of]--> Subrealm --[is_direct_region_part_of]--> Rea

                                                                  FILTER ( !=

  Neighbor --[is_direct_region_part_of]--> Subrealm --[is_direct_region_part_of
```

1. Retrieves NA15's geometry and walks up its hierarchy to find its realm.
2. Finds all touching bioregions (same `geof:sfTouches` pattern as before).
3. For each neighbor, walks up the hierarchy to find its realm.
4. Filters to keep only neighbors whose realm differs from NA15's realm.

Would you like me to store this query in the Query Library?

You

Yes, please save that query and the previous one as well if possible

GraphTalker

Both queries are now saved in the Query Library:

1. **"Find bioregions sharing a boundary with a given bioregion"** -- the spatial neighbor query using `geof:sfTouches`
2. **"Find boundary neighbors belonging to a different realm"** -- the cross-realm neighbor query combining spatial adjacency with hierarchy comparison

You can find them both in the Query Library tab.

You can find them both in the Query Library tab.

You

Which bioregion, subrealm, and realm contain longitude −0.1276 and latitude 51.5072? Use the stored geometry and region relationships.

GraphTalker

The point at longitude **-0.1276**, latitude **51.5072** (London, UK) falls within:

Level	Name
Bioregion	Great Britain Ireland And Faroe Islands (PA9)
Subrealm	Anglo Celtic Isles
Realm	Western Eurasia

This used the same stored query pattern -- `geof:sfContains` for the spatial test, then two hops up via `oeb:is_direct_region_part_of` through the Subrealm to the Realm.

You

Does any bioregion geometry in this repository contain longitude −140 and latitude 0? If none does, report that no matching geometry was found.

GraphTalker

No matching geometry was found. No bioregion boundary in this repository contains the point at longitude **-140**, latitude **0**.

This makes geographic sense -- that location is in the open **eastern Pacific Ocean**, roughly 1,500 km west of the Galapagos Islands. The One Earth bioregions dataset covers terrestrial (land-based) regions, so oceanic points far from any landmass will not fall within any stored bioregion polygon.